

Quantum Molecular Computing: The Self-Assembly Model

MICHAEL CONRAD

Department of Computer Science, Wayne State University, Detroit, Michigan 48202

Abstract

The principle of macromolecular self-assembly is used to construct a model of computing that exploits quantum effects to achieve enhanced real-time capabilities. Signals impinging on a device (or biological cell) trigger the appearance of macromolecules that self-assemble into a mosaic. Adaptor enzymes recognize features of the mosaic and link these to the output of the device. In this way, a symbolic pattern recognition problem is converted to a free-energy minimization process. A Hartree-type self-consistent-field formalism is developed for treating the self-assembly process. The formalism demonstrates that the parallelism inherent in the quantum mechanical wave function (the superposition of electronic states) can speed up the exploration of the potential surface, thereby increasing computational search power over what can be achieved with conventional models of computation. © 1992 John Wiley & Sons, Inc.

1. Introduction

Biological pattern formation at the subcellular level is largely mediated by the self-assembly of macromolecular structures. One goal of the present paper is to show that self-assembly can also subserve cellular pattern recognition and that it is possible to construct networks of cellular pattern recognizers capable of computing any function computable by conventional (digital) computers. The second goal is to show that quantum effects that speed up the self-assembly process allow the real-time capabilities of the self-assembly mode of computing to exceed those of conventional computers for important classes of problems.

The speed-up effect derives from the fact that self-assembly is an energy-minimization process. Digital computers, of course, use energy, but energy plays no role in controlling the course of the computation. The chief feature of the self-assembly mode of computing is that the sequence of states is controlled by an energy landscape. Solution of a problem in these models corresponds to finding a suitable free-energy minimum. The speed-up effect has its origin in the fact that systems that are sufficiently microscopic can exploit the superposition of quantum mechanical states ("quantum parallelism") to explore the potential surface.

Macromolecular systems are uniquely suited to this kind of search because they allow for electronic-conformational interactions. The minimum-energy state is determined primarily by the complementary shape properties of interacting molecules; hence, by the nuclear conformations. Our assumption is that the docking of the molecules is influenced by the electronic wave function and, therefore, is sped up

by the superposition of electronic states. The non-Born–Oppenheimer character of the complex is thus essential to the model. The complexity of the interaction is too great to allow for a detailed calculation. It is possible, however, to construct a self-consistent theory-type formalism that incorporates the required interactions in a general way and to use this to elicit some of the general conditions under which the speed-up effect occurs.

2. Computing via Self-assembly

The basic processing unit could be either a biological cell or a device (as yet hypothetical). In either case it will be called a transduction–amplification (TA) module since it transforms input patterns to output patterns by transducing the former to the molecular level, processing them at that level and then amplifying them to a macroscopic output signal or action (Figure 1). The sequence of operations

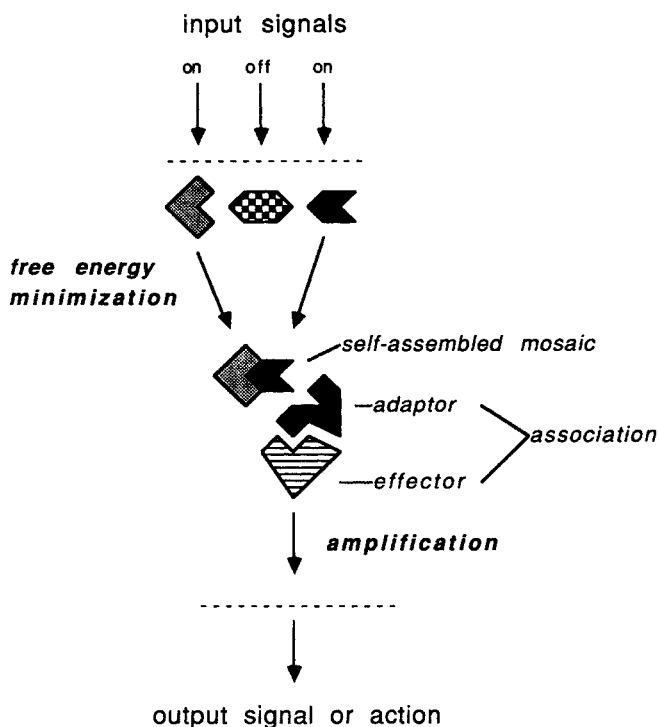


Figure 1. Transduction–amplification module operating on the basis of the self-assembly principle. Signals arriving along the different input lines release differently shaped macro-molecules. Consequently, different patterns of input signals yield polymacromolecular mosaics with distinctive shape features. Adaptor enzymes pick out shape features associated with different subsets of input patterns and link these to the output of the cell (or device) by activating specific effector proteins. The adaptor enzyme in the diagram is depicted as ready to complex with the mosaic, and the effector, as ready to complex with the adaptor.

is **input signal pattern** → **released macromolecules** → **self-assembled mosaic** → **readout adaptor** → **output**. Each input signal line triggers the release of a specifically shaped macromolecule. The signal pattern impinging on the TA module is thus represented as a set of conformations. These self-assemble into a mosaic, somewhat in the fashion of jigsaw puzzle pieces. The mosaic will have numerous shape features, each associated with common features of different input patterns. Adaptor enzymes connect these shape features to effector proteins that control output (e.g., membrane electric activity in a cell). The number of different groups of input signal patterns that the cell can recognize depends on the number of different types of adaptor enzymes.

The advantage of the self-assembly model of computing is that it converts a symbolic pattern recognition problem into a free-energy minimization process. This is due to the representation of the input signals as specific molecular shapes. The time required for self-assembly of molecules in close proximity might be on the order of milliseconds, or less if the input signals were only required to trigger conformational rearrangements within an existing complex. However, the real-time capability of the TA module should be measured in terms of the number of digital switching operations required to duplicate the input–output transform that it performs [1]. This would be enormous if even a small fraction of the number of digital operations required for a reasonably accurate simulation of the self-assembly dynamics within the module are relevant to duplicating this transform.

TA modules operating on the principle of self-assembly are well suited to classifying input signal patterns that have a highly variable spatial or temporal structure. Spatially distorted patterns will give rise to the same self-assembled mosaic as long as the self-assembling molecules remain in active conformations long enough to come together at a common point in space. Temporal variations will also yield the same mosaic as long as all the self-assembling molecules are active at some common point in time. The number of spatiotemporal variations on an input pattern grows in a combinatorially explosive way. Consequently, problems of this type are extremely computation intensive for conventional computers and generally intractable. Finding common features of topologically different input patterns is also extremely difficult. But as noted above, the repertoire of adaptor enzymes will group topologically different patterns together as long as these are represented by common shape features of the self-assembled complex.

Networks of TA modules can implement any function computable by a conventional digital computer. This follows trivially from the fact that it is possible to implement any such function with simple threshold elements, i.e., modules that fire if the sum of their inputs equal or exceed a threshold [2]. Threshold elements of this type, known as McCulloch–Pitts formal neurons, are special cases of TA modules that perform averaging operations on the input signal patterns. The TA networks are more efficient since TA modules can respond to the detailed structure of the input patterns. Many simple threshold elements are, in general, required to duplicate the input–output transform of a single TA module, and a single appropriately designed TA module might perform the same input–output transform as a network of simple threshold elements. Networks of simple threshold elements

are potentially computation universal (i.e., interpretively programmable) and, consequently, networks of TA modules are potentially computation universal as well.

TA modules operating on the basis of the self-assembly principle cannot be programmed to perform desired functions in the same manner that networks of threshold elements can, i.e., with a simple users manual. "Programming" means "engineering" the molecules that participate in mosaic formation and "engineering" the adaptor molecules. If this could be done with a simple users manual, it would mean that interacting molecules would have to be organized in such a special manner that the principles and equations of quantum chemistry become irrelevant to all their essential properties. This is implausible and certainly not the case for biological macromolecules. The manner of programming must be through learning or adaptive procedures.

Two important types of adaptation processes are available. The first is evolution through variation and selection. The technological analog is directed evolution (or protein engineering). Natural or directed evolution produces the repertoire of mosaic-forming molecules and adaptors. The second "self-programming" process is adaptive self-stabilization of different initial organizations of this evolved repertoire. Adaptive self-stabilization operates through an error signal. If the input-output transform of the TA module is incorrect, the internal structure of the module is disturbed, causing the locations of the binding sites to which the mosaic forming and adaptor molecules are initially attached to be altered. This process continues until the TA module converges on the correct output. In a biological cell, this would be a homeostasis conferring output. The self-stabilization mechanism requires some kind of a substructure (such as the cytoskeleton) that exhibits hysteresis (i.e., remains in the altered form induced by the error signal). After the TA module recognizes an input pattern, a change in the internal milieu must occur that allows the mosaic-adaptor complex to disassemble. The molecules that contribute to complex formation then diffuse (or are energetically driven) back to their original binding sites.

Many variations on the self-assembly model are possible. For example, a number of investigators have proposed that the membrane and cellular cytoskeletons serve as a fast signal transmission system [3-6]. Such a network could rapidly carry information from widely separated regions of the cell to mosaic-adaptor complexes that process them via the self-assembly-adaptor mechanism and that then trigger further intracellular signals. Signaling might be mediated by conformation changes, mechanical motions of the cytoskeletal elements, or vibratory dynamics of the cytoskeletal network. A vast intracellular network with self-assembling nodes is possible. It is only necessary that the mosaic at each node undergo rearrangements that at least transiently exhibit shape features recognizable by enzymes.

3. Electronic-Conformational Interactions

The image of molecular shapes self-assembling like pieces of a jigsaw puzzle is metaphorical. The chances that two macroscopic jigsaw puzzle pieces would ever find a complementary fit through random bombardment is negligible. In reality, the atomic nuclei undergo significant dynamic motions relative to other nuclei in

the same molecule in the process of complex formation. Under these conditions, the interaction between the electronic and nuclear wave functions becomes important [7]. It is this interaction that allows the superposition of electronic states to speed up the docking process and, thereby, to speed up the input-output transform performed by the TA module over and above what could be understood on the basis of a purely classical model and over and above what could be obtained with a computer using comparable material resources in a classical mode.

The energy-minimization problem is schematically captured in the two potential surfaces schematically illustrated in Figure 2. This is a two-dimensional projection of a generally very high-dimensional pair of surfaces, with one surface corresponding to the potential energy of the electronic system and one corresponding to the potential energy of the nuclear system. In reality, these potentials are dependent on each other because they share dependencies on some of the same coordinates. However, it is, in principle, possible to assign a potential energy to the electronic system relative to its lowest-energy state for the given collection of assembling molecules (under the assumption that the covalent bond structure is not altered in the course of complex formation). Similarly, it is, in principle, possible to assign a potential energy to the nuclear conformation relative to its lowest possible energy state. Free-energy surfaces with a perhaps slightly different form would correspond to these notational potential energy surfaces. We will ignore the entropy contribution to complex formation, however, since this complication would obscure the role of the superposition principle. Also, since complex formation is a form of crystallization, dominated by the specific fit of the participating molecules, it is reasonable to suppose that energy dominates entropy.

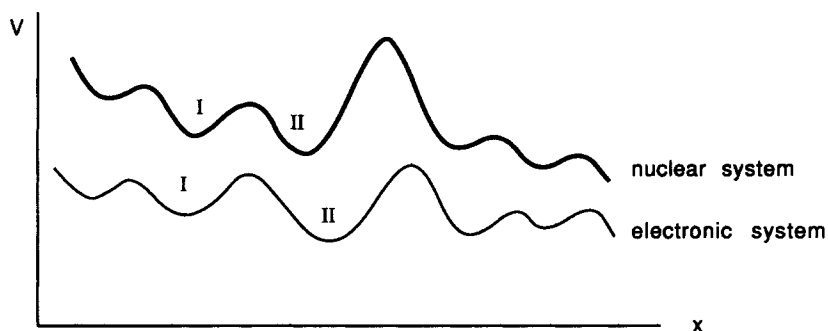


Figure 2. Schematic one-dimensional projection of nuclear and electronic potential surfaces. The surfaces are coupled since they share dependencies on both electronic and nuclear coordinates (denoted by x_e and x_n in the text), but the set of coordinates on which they significantly depend is not, in general, the same. Corresponding minima of the two surfaces (denoted by the same Roman numerals) are for pictorial convenience, depicted as having corresponding x coordinates, but this would not, in general, be the case. Instead, corresponding coordinate values are linked through conformation. Thus, minimum I of the electronic surface is the lowest-energy state of the electronic system when the nuclear system is in conformation I, and minimum I of the nuclear surface is the lowest energy state of the nuclear system when it is in conformation I.

It is not possible to tell from a two-dimensional projection of the type illustrated in Figure 2 whether the electronic or nuclear systems could fall downward from one of the initial to one of the final stage locations without being caught in some local minimum. A purely downhill pathway could pass through the unrepresented dimensions. However, it is inevitable that the downward progress would be impeded by numerous local minima because irregularly structured surfaces usually cannot slide smoothly against one another. This would be true even for well-engineered keys and locks if the former had to start from randomly chosen contact points. It is incorrect, of course, to extrapolate the frictional interactions between macroscopic objects to the molecular level. However, the self-assembled complex is held together by weak interactions, especially the short-range van der Waal's interactions, of the type that are responsible for macroscopic friction. Many partial complementary fits are possible, and these would each correspond to a local minimum.

The assumption that the electronic and nuclear wave functions are coupled means that the two potential surfaces are coupled. *The physical picture that will emerge from the formal model to be developed is that the nuclei of the self-assembling molecules exhibit a bobbing motion in response to the changing phase relations among the different components of the electronic wave function, over and above the motions connected with thermal agitation. Interference effects among different components of the wave function then serve to trigger the transition from one nuclear conformation to another.* The resulting changes in the nuclear potential lock the electronic wave function into a form that is more likely to be associated with the new than with the old conformation. Self-assembly in this picture is mediated by multiple cycles of this type.

4. Self-Consistency Model

The cycle of electronic-conformational interactions previewed above is in some respects reminiscent of Hartree-Fock-type self-consistent-field theories. The difference is that the recursive interaction between wave functions and potentials that leads to self-consistency is viewed as reflecting the self-organizing dynamics of the complex and not as an approximation scheme with no in-principle connection to the manner in which atomic and molecular structure arises. The justification is that the self-assembling complex should in the background be governed by a true potential that is time-independent. By itself, this true potential is not useful though. The wave function connected to it would contain too much information, including information about the heat bath (since complex formation involves energy dissipation). If this extraneous information is discarded, the potential becomes time-dependent. As self-assembly proceeds, the relevant potential continues to change. The wave function is controlled by the potential changes, leading to the cyclic approach to self-assembly. This is the conceptual justification for interpreting the recursive interaction between wave function and potentials as having real physical significance.

The objective of this and the following section is to develop a general scheme of this type that can serve to demonstrate in a conceptual but explicit manner the

role of the superposition principle in speeding up complex formation. For this purpose, it is convenient to think in terms of two notational wave functions, one for the nuclear system and one for the electronic system, without attempting to include detailed information about electronic structure, nuclear configuration, and hydrogen bonds. These factors are obviously all important, but including them would obscure the basic structure of the interaction. The interaction is obviously highly nonlinear. For the purpose of making the discussion as explicit as possible, the nonlinearity will be concentrated at the points of updating the potentials and wave functions. Elsewhere, the scheme will be linearized as much as possible, so as to utilize time-dependent perturbation theory.

The wave function of the whole complex is a function of both the electronic and nuclear coordinates. It can be written (notationally) as a solution of the Schrödinger equation

$$\begin{aligned}
 -\sum_r \frac{\hbar^2}{2m_r} \nabla_r^2 \Psi(x_e, x_n, t) - \sum_s \frac{\hbar^2}{2M_s} \nabla_s^2 \Psi(x_e, x_n, t) + V(x_e, x_n, t) \Psi(x_e, x_n, t) \\
 = i\hbar \frac{\partial \Psi(x_e, x_n, t)}{\partial t}, \quad (1)
 \end{aligned}$$

where the m_r refer to the electrons, the M_s refer to the nuclei, x_e represents all the relevant electronic coordinates, x_n represents the relevant nuclear coordinates, and t is time. Before the molecules interact, the wave function of the joint system splits into a product of wave functions of the individual molecules. When the molecules make contact, the wave functions may still be viewed as independent, but modified by an interaction term that can be treated as a perturbation and that changes as self-assembly proceeds.

To exhibit the role of the superposition principle, however, it is not necessary to explicitly represent the above splitting. It is sufficient and much clearer to focus on the interaction of the electronic and nuclear subsystems of the complex taken as a whole. If the Hamiltonian of the complex could be decomposed into a sum of Hamiltonians that depend separately on its electronic and nuclear coordinates, it would be possible to write

$$\Psi(x_e, x_n, t) = \Psi_e(x_e, t) \Psi_n(x_n, t), \quad (2)$$

where $\Psi_e(x_e, t)$ is the electronic wave function of the complex and $\Psi_n(x_n, t)$ is its nuclear wave function. For this splitting to be a useful approximation, it is necessary for the potential energies of the electronic and nuclear systems to be approximately additive and, therefore, to depend on different coordinates. In a system undergoing significant intermolecular and intramolecular nuclear reorientations, the assumption of separable coordinates is untenable. The potential controlling the behavior of the electrons will depend significantly on the nuclear coordinates and also on interactions among the electrons. Similarly, the potential controlling the behavior of the nuclei will depend significantly on electrons in molecular orbitals, on the interactions among the nuclei, on surface interactions and distortions, and on hydrogen bonds.

The time scales of electronic and nuclear configurational subsystems are so different, however, that it can be instructive to consider the extremely idealized case in which they are describable by wave functions that are independently updated and that become correlated through mutually stabilizing interactions. The idealization here is that the potentials do not significantly depend on any of the same coordinates at the same time during the linearizable portions of the time development. The correlation of the wave functions is concentrated into the time points at which the potentials are updated. Apart from these points, the potential function can then with reasonable approximation be split into two parts:

$$V(x_e, x_n, t) = V_n(x_e, x_n, t) + V_e(x_e, x_n, t). \quad (3)$$

Here, V_n is the potential controlling the behavior of the electronic system, V_e is the potential controlling the behavior of the nuclear system, and the temporal separation of coordinate dependencies is not at this point explicitly specified. The reversal of subscripts relative to those of the corresponding wave functions (e.g., V_n is the potential for Ψ_e) is intended to indicate that the nuclei make a more significant contribution to the potential governing the time evolution of the electronic system than do interactions among the electrons and that the electronic system makes a bigger contribution to the potential governing the time evolution of the nuclear configuration than do the interactions among the nuclei.

The above idealization will serve to provide a lower bound on the importance of electronic-conformational interactions in complex formation. If the superposition of electronic states plays a role under the extreme condition that the nuclear and electronic systems do not share significant coordinate dependencies at the same time, it would presumably play an even more important role when this restriction is relaxed. With this understanding, the Schrödinger equation for the whole complex [Eq. (1)] separates into two equations:

$$-\sum_r \frac{\hbar^2}{2m_r} \nabla_r^2 \Psi_e(x_e, t) + V_n(x_e, x_n, t) \Psi_e(x_e, t) = i\hbar \frac{\partial \Psi_e(x_e, t)}{\partial t} \quad (4a)$$

$$-\sum_s \frac{\hbar^2}{2M_s} \nabla_s^2 \Psi_n(x_n, t) + V_e(x_e, x_n, t) \Psi_n(x_n, t) = i\hbar \frac{\partial \Psi_n(x_n, t)}{\partial t}. \quad (4b)$$

To reexpress these equations in terms of a self-consistency scheme, we suppose that their solutions can be expressed in terms of a series of interaction cycles, each of which can be divided into a linearizable part and a highly nonlinear updating part. During the linearizable part, $\Psi_e(x_e, t)$ and $\Psi_n(x_n, t)$ evolve. During the nonlinear part, the potentials, V_e and V_n , are updated to reflect the changes in $\Psi_e(x_e, t)$ and $\Psi_n(x_n, t)$.

The updating process may be expressed notationally by writing

$$V_e(x_e, x_n, t) = V_e(\Psi_e(x_e, t), \Psi_n(x_n, t)) \quad (5a)$$

$$V_n(x_e, x_n, t) = V_n(\Psi_e(x_e, t), \Psi_n(x_n, t)), \quad (5b)$$

where the functional notation expresses the fact that V_e and V_n depend on features of both the electronic and nuclear systems. The updated functional forms are not directly determined by the Schrödinger equations (4a) and (4b). This is because the potentials are classical quantities. In principle, it would be necessary to construct operators to obtain them from the wave functions. Which operators should be used is not in itself determined by the equations of motion. In practice, auxiliary considerations would have to be utilized.

The introduction of operators does *not* mean that measurement plays a role in the self-consistency scheme any more than the calculation of updated potentials from electron densities means that measurement plays a role in the Hartree–Fock treatment of a multielectron atom. It does, however, have a physical interpretation in the present scheme. As stated earlier, the updating of the potential means that details of the true potential are being discarded because different features of it become relevant as the relative orientation and internal molecular structure of the self-assembling molecules change. This discarding of information has an objective physical basis in the interaction of the complex with the heat bath. To describe this interaction, it would be necessary to move to a statistical description. However, this would still not alter the requirement for introducing operators. In a simple physical system, the operators could represent simple averaging procedures. In an intricate system, such as a self-assembling complex, it is not obvious a priori what specific details should be averaged away. To use the scheme to be developed for actual calculations, it would be necessary to introduce some workable procedures for extracting potentials from the wave functions. However, for the present conceptual purposes, it is not necessary to specify these procedures (or the operators corresponding to them).

5. The Iteration Scheme

The wave functions on which the updating of V_e and V_n depend [in Eqs. (5a) and (5b)] are themselves solutions of Schrödinger equations (4a) and (4b), governed by the pre-updated forms of V_e and V_n . To express this dependence, and also to make explicit the time intervals over which the potentials are valid, we will use the notation

$$\Psi(x, t + \tau_b) = \Psi(V(x, t + \tau_a), t + \tau_b). \quad (6)$$

Here, $\Psi(x, t + \tau_b)$ is the (electronic or nuclear) wave function at time $t + \tau_b$, and $\tau_b - \tau_a$ is the time interval over which the (nuclear or electronic) potential controlling the time evolution of this wave function remains applicable.

The assumption is that time-dependent perturbation theory is adequate for dealing with any changes in the potential functions during the intervals. The potentials then take the form

$$V(x, t) = V'(x) + v(x, t) \quad (7)$$

in a piecewise continuous manner, where $V'(x)$ is a time-independent unperturbed potential applicable during the interval $\tau_b - \tau_a$, and $v(x, t)$ is a small time-dependent

perturbation that develops over this interval. The solution of the Schrödinger equation for $V'(x)$ may then be separated into time-independent and time-dependent components,

$$\Psi'(x, t) = \psi(x)\phi(t), \quad (8)$$

in a piecewise continuous manner. This assumption will simplify the presentation, but because it will not be necessary to actually solve the Schrödinger equation for the unperturbed potentials, it is more restrictive than necessary.

Let the time interval $t_{u+1} - t_u$ denote the u th cycle of nuclear-electronic interactions. The electronic system undergoes an evolution over a time interval $\tau_{e1(u)}$ and the nuclear system follows over a time interval $\tau_{n(u)} - \tau_{e1(u)}$. The cycle is completed when the electronic system makes a fast adjustment to the potential defined by the new nuclear wave function over a time interval $\tau_{e2(u)} - \tau_{n(u)}$. The subscript (u) associates these intervals with the start time of the cycle over which they are valid.

Iteration of such cycles should eventually lead to a state in which the relationship between the wave functions and potentials is self-consistent. The self-consistent state might correspond to the self-assembled complex or to some dissociated organization. In either case, the iteration scheme may be written as

$$\begin{aligned} \Psi_e(x_e, t_u + \tau_{e1(u)}) &= \Psi_e(V_n(x_e, x_n, t_u), t_u + \tau_{e1(u)}) \\ &= \Psi_e(V_n(\Psi_e(x_e, t_u), \Psi_n(x_n, t_u)), t_u + \tau_{e1(u)}) \end{aligned} \quad (9a)$$

$$\begin{aligned} \Psi_n(x_n, t_u + \tau_{n(u)}) &= \Psi_n(V_e(x_e, x_n, t_u + \tau_{e1(u)}), t_u + \tau_{n(u)}) \\ &= \Psi_n(V_e(\Psi_e(x_e, t_u + \tau_{e1(u)}), \Psi_n(x_n, t_u)), t_u + \tau_{n(u)}) \end{aligned} \quad (9b)$$

$$\begin{aligned} \Psi_e(x_e, t_u + \tau_{n(u)} + \tau_{e2(u)}) &= \Psi_e(V_n(x_e, x_n, t_u + \tau_{n(u)}), t_u + \tau_{n(u)} + \tau_{e2(u)}) \\ &= \Psi_e(V_n(\Psi_e(x_e, t_u + \tau_{n(u)}), \Psi_n(x_n, t_u + \tau_{n(u)})), t_u + \tau_{n(u)} + \tau_{e2(u)}). \end{aligned} \quad (9c)$$

Here, $t_{u+1} = t_u + \tau_{n(u)} + \tau_{e2(u)}$ and t_0 can be taken as the time of initial contact. The nuclear wave function $\Psi_n(x_n, t_u)$ is used in Eq. (9b) rather than $\Psi_n(x_n, t_u + \tau_{e1(u)})$ since $\tau_{n(u)} > \tau_{e1(u)}$, but actually it would be more precise to use the latter. Similarly, $\Psi_n(x_n, t_u + \tau_{n(u)})$ could be used for $\Psi_n(x_n, t_u + \tau_{n(u)} + \tau_{e2(u)})$ when updating V_n (and setting $u = u + 1$) for reentering Eq. (9a).

Each of the wave functions in the scheme, and, hence, each of the potentials, is defined in terms of the initial electronic and nuclear wave functions, $\Psi_e(x_e, t_0)$ and $\Psi_n(x_n, t_0)$ or, alternatively, in terms of the initial nuclear potential, $V_n(x_e, x_n, t_0)$, and an initial electronic potential, $V_e(x_e, x_n, t_0 + \tau_{e1(0)})$. The nuclear potential controls the time evolution of the electronic wave function, as described by the Schrödinger equation for the electronic system [Eq. (4a)], and the resulting electronic potential then controls the time evolution of the nuclear wave function, as described by the Schrödinger equation for the nuclear system [Eq. (4b)]. The nuclear potential implicit in these updated wave functions controls the shorter time scale ($\tau_{e2(u)}$) adjustment of the electronic wave function. The wave functions are again updated and V_n recalculated. At this point, the time index is updated ($t_1 = t_0 + \tau_{n(0)} +$

$\tau_{e2(0)})$ and the process repeated for times t_1, t_2, t_3 , and so forth, until the system stabilizes in the final self-assembled form or dissociates.

When V_e is operative, the electronic wave function is assumed to be static. When V_n is operative, the nuclear wave function is static. This establishes the continuity conditions. The recalculation of V_e and V_n includes the updating of the boundary conditions (viewed as built into their specific functional forms). The time intervals $\tau_{e1(u)}$, $\tau_n(u)$, and $\tau_{e2(u)}$ could, in principle, be determined by solving the applicable equation [(9a), (9b), or (9c)] with a trial time interval Δt . If an updated wave function (Ψ_e or Ψ_n) yields the same potential function (V_n or V_e) as the original wave function, the time is incremented by Δt and the equation is solved again. Instead of recalculating the potentials at each step, it may be possible to make some reasonable assumptions about how much the probability densities, $p(x_e, t) = \Psi_e^*(x_e, t)\Psi_e(x_e, t)$ and $p(x_n, t) = \Psi_n^*(x_n, t)\Psi_n(x_n, t)$, must change before the potential changes. The change in the potential function marks the end of the time period over which it is applicable. Control is then shifted to the next equation in the scheme.

The conditions for a stable configuration are

$$\Psi_e(x_e, t_{u+p}) = \Psi_e(x_e, t_u) \quad (10a)$$

$$\Psi_n(x_n, t_{u+p}) = \Psi_n(x_n, t_u), \quad (10b)$$

where $p \geq 1$. If $p \rightarrow \infty$, then the system is strongly stable. If p is finite (or at least not very large), the system is metastable. If the complex dissociates $\Psi_e(x_e, t_u)$ and $\Psi_n(x_n, t_u)$ both split into separate wave functions for the individual molecules. The wave functions might change gradually over the linear region, but, nevertheless, yield a nonnegligible change in the potential functions after significant time evolution occurs. Or they may be metastable over the linear region, automatically yielding the same potential functions, but undergoing transitions to new metastable states with some probability.

We can now reexamine our assumptions in the light of the formalized scheme. The potentials are updated at different points in time, due to the different time scales of nuclear and electronic processes. The cross dependencies are irrelevant for the time intervals over which these potentials retain validity because only one of the wave functions evolves over any given interval of time. When this is used to update the potential governing the other wave function, the latter must be recalculated. All the cross-dependencies, as noted earlier, are concentrated into the highly nonlinear dynamics of updating. The approximation of separable wave functions breaks down at the updating points, since this is where the correlation process is concentrated.

6. Role of the Superposition Principle

To demonstrate the speed-up effect, it will be sufficient to consider a jump between two neighboring conformations, to be denoted by I and II, each of which is a local minimum of both the electronic and nuclear potentials. To keep matters as simple

as possible, we can suppose that the potential barriers separating these minima (also to be denoted by I and II) from other regions of the potential surface are infinitely high. Under these conditions, the coupled potential surface illustrated in Figure 2 reduces to a coupled double-well potential. Also, for simplicity, we will ignore the fact that each minimum might, in fact, comprise a family of subminima separated by low barriers [8].

According to the superposition principle, it is possible to expand Ψ_e and Ψ_n in terms of complete sets of basic functions:

$$\Psi_e(x_e, t) = \sum_r a_r(t) e_r(x_e) \exp\left(\frac{-iE_{e(r)}t}{\hbar}\right) \quad (11a)$$

$$\Psi_n(x_n, t) = \sum_s b_s(t) n_s(x_n) \exp\left(\frac{-iE_{n(s)}t}{\hbar}\right). \quad (11b)$$

Here, the $e_r(x_e)$ are a complete set of basic functions for the electronic system, the $n_s(x_n)$ is a complete set of basic functions for the nuclear system, and the a_r and b_s are expansion coefficients. The $E_{e(r)}$ and the $E_{n(s)}$ are the energies associated with the e_r and n_s , respectively.

Since the eigenfunctions range over all possible states, some combinations of them should be associated with minimum I and some with minimum II. In principle, we could take any complete set of basic functions for this purpose. But to associate both the electronic and nuclear eigenfunctions with minimum I and II in as unique a way as possible, it is most convenient to suppose that we are working with the eigenfunctions of the true potential—in the present case, the potential function describing two coupled double wells (plus a time-dependent perturbation to represent the interaction with the radiation field). When the potentials corresponding to I are applicable, the eigenfunctions associated with I are dominant; when the potentials corresponding to II become applicable, the eigenfunctions associated with II dominate. The set should also include eigenfunctions that cannot unambiguously be associated with either I or II. But, initially, these would also have small expansion coefficients. The eigenfunctions constructed for these potentials taken in isolation may be viewed as approximations to the relevant eigenfunctions that, in principle, could be constructed from the true potential. Alternatively, we could, in principle, assume that we know the true potential but continually discard features of it as complex formation proceeds. In practice, of course, it is unfeasible to actually construct the eigenfunctions of the true potential, except by using the local potentials to approximate them. But they exist in principle and, consequently, it is legitimate to work with them notationally for the purposes of constructing a proof.

Expanding the wave functions in terms of initial sets of eigenfunctions allows the time dependence of the potentials, both in the linear region and at the updating points, to be represented in the time dependence of the expansion coefficients. Dividing the potential function at each stage into a time-independent and a time-dependent part allows $\Psi_e(x_e, t)$ and $\Psi_n(x_n, t)$ to be expanded in terms of unperturbed wave functions with stationary and vibratory parts. The latter oscillate with a fre-

quency $v_r = E_{e(r)}/\hbar$ and $v_s = E_{r(s)}/\hbar$. This oscillatory time dependence is distinct from the time dependence expressed in the expansion coefficients. The changing phase relations among these vibratory terms controls the manner in which the different components of the wave function contribute to the time development of the system [9].

Choosing approximation assumptions that elicit the vibratory terms in an explicit form thus makes it possible to show that under reasonable conditions the interference terms contribute to self-assembly. The key point is to utilize the natural associations between the eigenfunctions of the true potential and the accessible minima (if arbitrary basic functions were used, the associations would have to be between superpositions and accessible minima). The naturalness of the association follows from the fact that there are only three possible classes to which each of the e_r and n_s can belong. Class I is associated with minimum I and class II with minimum II, either as ground or excited states of these minima. Class III cannot be unambiguously associated with either minimum. The expansions then reduce to

$$\begin{aligned} \Psi_e(x_e, t) = & \sum_{rel} a_I(t) e_I(x_e) \exp\left(\frac{-iE_{e(I)}t}{\hbar}\right) + \sum_{rell} a_{II}(t) e_{II}(x_e) \exp\left(\frac{-iE_{e(II)}t}{\hbar}\right) \\ & + \sum_{relll} a_{III}(t) e_{III}(x_e) \exp\left(\frac{-iE_{e(III)}t}{\hbar}\right) \end{aligned} \quad (12a)$$

$$\begin{aligned} \Psi_n(x_n, t) = & \sum_{sel} b_I(t) n_I(x_n) \exp\left(\frac{-iE_{n(I)}t}{\hbar}\right) + \sum_{sell} b_{II}(t) n_{II}(x_n) \exp\left(\frac{-iE_{n(II)}t}{\hbar}\right) \\ & + \sum_{selll} b_{III}(t) n_{III}(x_n) \exp\left(\frac{-iE_{n(III)}t}{\hbar}\right), \end{aligned} \quad (12b)$$

where I, II, and III are used to denote the index set associated with the respective classes.

In principle, these expansions mean that the complex cannot, in general, be assigned with certainty to minimum I or II (or to an ambiguous state) at any given time, since it is, in general, not in a true stationary state. However, this indefiniteness is of negligible significance for the nuclear system taken in isolation, because the de Broglie wavelengths of the atomic nuclei are extremely small in comparison to the shifts in nuclear position that would accompany the transition between distinct conformations (corresponding to minimum I and II). Thus, if the complex is in conformation I at time t_u , we can take $b_{II}(t_u) \ll b_I(t_u)$ for all values of the indices I and II, and, similarly, we can assume that the $b_{III}(t_u)$ are negligible. The important point is that even if the nuclear system is in a superposition of states the average values of the energies and position peak vary sharply. The interference terms between eigenfunctions belonging to I therefore do not play an important role. The overlap between the wave function and eigenfunctions belonging to II is negligible and, apart from thermal agitation, should remain negligible. If we generalized to density matrices, this would mean that the mixture of wave functions would almost solely

come from an equivalence class of wave functions associated with either I or II, but not with both.

The situation is quite different for the electronic system. The de Broglie wavelengths of electrons are comparable to or greater than the internuclear spacings. The average values of the positions do not, in general, peak sharply and may in some cases be significantly delocalized. It is not possible to think in terms of definite shape features, and, therefore, the concept of conformation is inappropriate to the electronic system taken in isolation. We can nevertheless reasonably assign the electronic eigenfunctions to conformations I or II if they are in at least a quasi-stationary state of the nuclear potentials associated with these conformations. Thus, if the nuclear system is initially in minimum I, one or more members of the $a_I(t_u)$ will initially be large and all of the $a_{II}(t_u)$ negligible.

For definiteness, suppose that the electronic system is initially in a single eigenstate of nuclear conformation I. The small time-dependent perturbation $v_n(x_e, x_n; t = t_u$ to $t_u + \tau_{eI(u)})$ will induce transitions to other eigenstates. If some of the eigenfunctions are degenerate, the wave function could change substantially, leading immediately to eigenfunctions in the ambiguous class or in class II. However, to strengthen the argument, let us restrict ourselves to the case most unfavorable for the transition, namely, that all of the perturbation-induced eigenstates belong to conformation I and that none of them fall into the ambiguous class or into class II. According to the superposition principle,

$$\Psi_e(x_e, t) = \sum_{rel} a_I(t) e_I(x_e) \exp\left(\frac{-iE_{e(I)}t}{\hbar}\right), \quad (13)$$

where $t = t_u + \tau_{eI(u)}$. It might initially be thought that perturbations with ultraviolet energies would, in general, be required to trigger transitions between different electronic energy levels in a biological molecule (since biological materials are colorless apart from pigments). This would be the case in molecules that satisfy the Born–Oppenheimer assumption that the nuclei can be treated as fixed relative to the electrons. This assumption completely breaks down for molecules assembling into a complex, and even for individual proteins, since intramolecular rearrangements of the nuclei are all important. Born–Oppenheimer allows the electronic energy to be added to the nuclear potential. Consequently, the infrared spectra is attributed to vibrations and rotations of the nuclei. But if the relative positions of the nuclei and electrons are not reasonably fixed, some of the charge acceleration, hence, some of the changes in infrared spectra in the course of self-assembly, should be attributed to the electronic system (to those electrons whose motions are not fixed relative to the nuclei).

Small, perturbative influences that are omnipresent due to the interaction with radiation, or due to both thermal and anharmonic motions of the nuclei, can then trigger transitions that contribute to the superposition of electronic states [Eq. (13)]. Using the interpretative postulate of quantum mechanics to convert this superposition to a probability density of electronic coordinates,

$$p(e, t) = \sum_r a_r^*(t) a_r(t) e_r^*(x_e) e_r(x_e) + \sum_r \sum_{r'} a_r^*(t) a_{r'}(t) e_r^*(x_e) e_{r'}(x_e) \exp\left[\frac{-i(E_{e(r)} - E_{e(r')}t)}{\hbar}\right], \quad (14)$$

where r and r' both belong to the conformation I index set and $r \neq r'$. Because of the interference between the states of different energy, the probability density changes with time. The electronic charge density must therefore be time-dependent in a manner that reflects the changing phase relationships among the different eigenfunctions. This means that V_e , the potential that controls the behavior of the nuclear system, also changes with time. It must be updated at time $t = t_u + \tau_{e1(u)}$, in accordance with the time evolution of Ψ_e given by Eq. (9a) and as required to solve Eq. (9b). The nuclear system will then undergo a time evolution, and, consequently, V_n , the potential that controls the behavior of the electronic system, must be updated at $t = t_u + \tau_{n(u)}$ [in accordance with Eq. (9b)]. This causes transitions of the electronic system to still other eigenstates, decreasing some of the previously large expansion coefficients and increasing some that were previously negligible. V_n is again updated when u is set to $u + 1$ and Eq. (9a) reentered, again inducing transitions to other eigenstates. The updated superposition may be written as

$$\Psi_e(x_e, t) = \sum_{rel} a_l(t) e_l(x_e) \exp\left(\frac{-iE_{e(l)}t}{\hbar}\right) + \sum_{re?} a_r(t) e_r(x_e) \exp\left(\frac{-iE_{e(?)t}}{\hbar}\right), \quad (15)$$

where $t = t_{u+1}$ and the question mark is used to represent the index set of the class of eigenfunctions with previously negligible expansion coefficients.

At this stage, the previously negligible eigenfunction (indexed by ?) could belong to class I, class II, or class III. However, at some stage, they will with near certainty be assignable to II, in accordance with three possible courses of development: fast divergence arising from mutual inconsistency of the electronic and nuclear wave functions, slow divergence due to such inconsistency, or fluctuation-induced transitions between self-consistent minima.

i. Fast Divergence

In this case, the expansion coefficients (a_l) of the electronic wave function change enough in one cycle (e.g., to include contributions from the a_{III} or even the a_{II}) to switch the nuclear conformation from I to II. The switching is induced by the time development of the electronic charge density, obtained from Eq. (15) at time $t = t_u + \tau_{e1(u)}$. The coefficients of the nuclear wave function thus switch from b_I to b_{II} at time $t = t_u + \tau_{n(u)}$. The nuclear conformation then drags the electronic system into the ground state associated with it, so that the electronic wave function becomes dominated by a_{II} at time $t = t_u + \tau_{n(u)} + \tau_{e2(u)}$. The physical picture is that perturbation-induced spreading out of the electronic wave function leads to interference effects that drag the nuclear system from conformation I to conformation II.

ii. *Slow Divergence*

The expansion coefficients (a_1) of the electronic wave function in this case cause the nuclear potential to change during the first cycle, but not enough to switch the conformation from I to II. The wave function given by Eq. (15) at time $t = t_u + \tau_{e1(u)}$ is still primarily a superposition of electronic eigenfunctions belonging to the a_1 , but more spread out within this class (as measured by an overlap integral). The consequent change in the electronic charge density alters $V_e(x_e, x_n, t_u + \tau_{e1(u)})$, which, in turn, alters the nuclear system eigenfunctions (but at this point still within the b_1 class). The potential V_n must be updated at time $t_u + \tau_{n(u)}$ and at time $t_u + \tau_{n(u)} + \tau_{e2(u)} = t_{u+1}$ [in accordance with Eq. (9c) and reentered Eq. (9a)]. This induces further spreading out of the electronic eigenfunctions, hence, addition of further perturbation to V_e and further alteration of the b_s . This divergent spreading out increases with each cycle, until the switch to the conformation II nuclear potential occurs. The physical picture is that the interference among the different components of the electronic wave function induces motions of the nuclei that amplify the interference effects, and that this cyclic process continues until the nuclear system switches to conformation II.

iii. *Fluctuation-induced Transitions*

Suppose that the electronic and nuclear wave functions converge to self-consistency in some local minimum. The complex will then satisfy the stability conditions specified in Eqs. (12a) and (12b). The iteration scheme may still be applied, but it will be driven purely by fluctuation rather than by inconsistency and fluctuation. A classical system would never escape this minimum except as a consequence of thermal fluctuation. The electronic wave function of a self-consistent complex will be a superposition of eigenfunctions whose expansion coefficients are spread out by perturbation. The spreading will reflect external perturbations and will not be amplified by the electronic-conformational interaction (as in the slow convergence process described above). The wave function, and therefore the probability density of the electronic coordinates, will evolve with time, due to the interference of eigenfunctions of different energy. Consequently, there is always a finite probability that the charge density will assume a form that either triggers a transition of the nuclear system to minimum II or that is inconsistent with the nuclear wave function and that therefore initiates a slow divergence process.

A newly occupied minimum (whether reached by fast or slow divergence, or through pure fluctuation) may itself be inconsistent. The complex will then jump to a new minimum or jump back to the old minimum. If minimum II has lower nuclear and electronic potential energies, the time spent in this minimum will, of course, be greater. If the final minimum is the global minimum and also self-consistent, it should be stable [in the sense of Eqs. (10a) and (10b)]. It will for all practical purposes be strongly stable if the free energy is significantly lower than any of the neighboring minima or if the barriers separating it from other minima are high. If it is not separated from neighboring minima by significant barriers, it will be metastable since it will eventually escape to the neighboring minima due to

inherent fluctuations. It is conceivable that neither the global nor any of the local minima are self-consistent. In this case, the complex will discover the lowest-energy minima and cycle among them.

7. Phenomenological Implications

The self-consistency formalism implies a number of qualitative features that could be used to assess its applicability to self-assembly and to related processes such as protein folding and DNA annealing.

i. Mediating Role of Hydrogen Bonds

The de Broglie wavelength of hydrogen is large enough to allow for tunneling motions relative to the bonding partners. Hydrogen bonds, in addition, appear to provide facilitating pathways for the tunneling movement of electrons in proteins [10]. The iteration scheme [Eqs. (9a)–(9c)] can be expanded to include these facts by treating the hydrogen bonds as a third system whose wave function and effect on the potentials is separately updated. The picture that emerges is that the interference effects among different components of the electronic wave function (especially involving the delocalized electrons) alters the hydrogen-bond structure. This alters the pathways for electron tunneling. The resulting modifications in the electronic wave function (which can be significant due to degeneracies) alters V_e , the potential that controls the time evolution of the larger nuclei. The resulting nuclear conformational change then further modifies and locks in the electronic wave function. These processes, and the interaction with electrons in different orbitals, would in reality be interlinked in a much more complex manner. The experimentally relevant point is that the above picture can be investigated by using deuterium and tritium to modify the tunneling probabilities of hydrogen bonds and by other chemical modification techniques.

ii. Effects of Electromagnetic Radiation

Brownian motion contributes to self-assembly by facilitating exploration of the potential surface through mutual shape exploration. In the present model, the exploration of the potential surface is also augmented by the spreading out of the electronic wave function (of those electrons not bound to atomic nuclei in a Born–Oppenheimer fashion). If the interactions between the electronic and nuclear wave functions are divergent, the spreading out will amplify, further facilitating the exploration of the potential surface. To induce nonrandom motions in the nuclei, the perturbing energies must be above kT (~ 0.026 eV). Absorption spectra associated with vibrations of the nuclei around some equilibrium position are typically on the order of tenths of an electron-volt and spectra associated with rotation of the molecules as a whole are typically on the order of hundredths of an electron-volt [11]. The spectral characteristics should change in the course of complex formation due to constraints on the nuclear motions. The self-consistency model requires that some of these changes be attributed to motions of the non-Born–Op-

penheimer electrons relative to the nuclei. Since very small energy shifts should be possible, and since such shifts can self-amplify through the electronic-conformational interaction, this would provide a possible means for sustained microwave radiation to affect biological systems. The model might therefore be tested by studying the effect of microwave radiation on rates of self-assembly. If proteins, lipids, and nucleic acids were colored (apart from chromophores), the dynamics of self-assembly would be sensitive to different frequencies of visible light. Conceivably, this feature could be studied by introducing light-absorbing moieties into proteins.

iii. Uniqueness of the Self-assembled State.

The self-consistency formalism implies that it is possible for global free-energy minimization, and, hence, for uniqueness of structure formation, to be a robust phenomenon for self-assembly, protein-folding, and DNA annealing. The requirement is that the molecules allow for significant electronic-nuclear interactions. Molecular structures could, of course, be designed to preclude such interactions or to utilize them to reach self-consistent local minima with extremely long residence times. The important point is that it is possible to design molecules that can undergo highly specific shape-based interactions and, nevertheless, not become trapped in local minima. Some milieu features that contribute to folding and self-assembly (such as chaperon molecules) may operate by introducing electronic-nuclear interactions that facilitate exploration of the potential surface.

8. Conclusions

The analysis in this paper demonstrates that it is possible to exploit the superposition principle to enhance computational search power. The basic idea has been to use the principle of self-assembly to construct a model of computation in which pattern recognition problems are mapped into the energy landscape of a self-assembling complex and then to build up more general computations from networks of pattern processing modules (the TA modules). The electronic wave function enhances the real-time pattern recognition capabilities of the modules by speeding up complex formation. Such speed-up is not possible in digital computing models, such as Turing machines or random access machines, since systems of this type execute formal computation processes that are precisely picturable in a classical sense. Quantum behavior, and, consequently, the superposition principle, can enhance the speed, reliability, size, and energy dissipation characteristics of the elementary switches used to implement such models. But all quantum features must be quenched so far as the number of time steps and processors required to perform pattern recognition tasks are concerned; otherwise, the picturable character of the models would be lost.

Computation via self-assembly is a special case of construction via self-assembly. The advantage of focusing on computation is that it is possible to characterize the capabilities of systems (including structure building systems) in terms of the number of time steps or material resources that some standard model of computation (such as Turing machines) would require to simulate these capabilities [12]. The analysis

presented in this paper demonstrates that the self-assembly mode of computing will exhibit substantial real-time advantages as compared to digital models for pattern processing problems and for problems that decompose into pattern processing problems.

Self-assembly and disassembly are ubiquitous inside the cell. So far, these processes have been viewed primarily from a pattern formation point of view rather than as a means of fast pattern recognition. The vast majority of macromolecular interactions are accompanied by conformational changes that are responsive to milieu influences that originate either from within the cell or from effects of the external environment on the cell. Whenever this occurs, the possibility exists that different macromolecules will be triggered by different specific influences and that the polymacromolecular conformations emerging from their interaction can be used to control the internally or externally directed actions of the cell in a manner appropriate to the context of influences. The implication is that the real-time functional capabilities of cells and organisms derive in significant measure from the nonpicturability of quantum phenomena and that many of these are therefore inherently beyond the reach of artificial intelligence models implemented on digital computers.

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